Name : [illegible]
Reg No : [illegible]
Course Code : CSA [illegible]
Course Name : Artificial Intelligence (AI)

1. Intelligent Agents in a Smart Home Automation System

Introduction.

An intelligent agent observes its environment and takes actions to achieve specific goals. In a smart home, it controls lighting, temperature, and security.

Types of Intelligent agents

1. Simple Reflex agent:

It takes action based on current inputs using predefined rules. For example, it turns on light when motion is detected. It is fast and simple but cannot learn from past experiences.

2. Model Based Agent:

It maintains informations about the environment and uses previous observations to make decisions. For example, it moitiors room occupancy and temperature. It handles changing situations but requires more memory.

3. Goal - Based Agent:

It selects actions to achieve a specific goal. For example, it adjusts the air conditioner to maintain the desired temperature. It supports flexible decision - making but requires planning.

4. Utility - Based agent:

It selects the best action by considering comfort, energy consumption, and security. For example, it reduces unnecessary electricity usage while maintaining comfort.

5. Best approach

A hybrid approach combining model - based, and utility based agents is most effective. It understands the environment, achieves goals, and balances comfort, energy efficiency and security.

Conclusion

intelligent agents make smart homes more comfortable, secure, and energy - efficient.

UCS and IDS in a Maze

Introduction

A robot in an unknown maze must find a path to exit without knowing the layout in advance. Uninformed search algorithms helps the robot explore possible paths.

Uniform Cost Search (UCS)

UCS expands the node with the lowest total path cost first.

Working:

* Start from the initial position.
* Explore the path with the lowest cost.
* Add neighboring positions to the priority queue.
* Continue until the exit is found.

Advantages:

- uses less memory.
- finds the shallowest solution when all all steps have equal costs.

Disadvantages:

Repeats some searches

May waste time exploring nodes again.

Time complexity: $O(b^d)$

Space complexity: $O(bd)$

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Relationship with Intelligent Agents

- Simple Reflex agents: Acts based on current Sensor inputs.
- Model - Based agent: Remembers visited positions and obstacles.
- Goal - Based agent: Searches for path to exit.
- Utility - Based agent: Considers path cost, energy, and safety.

Best approach

A model-based, goal-based agent using UCS is suitable when different paths have different costs. IDS is better when all steps have equal costs and memory is limited.

Conclusion:

UCS finds the least-cost path, where as IDS uses less memory. the choice depends on Path costs, maze size, and available memory.